

# Dual-Certified Sparse Inference for PSSA: Error-Bounded Memory Retrieval and Vocabulary Projection

Architecture Research Contribution

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## Abstract

Plastic State-Space Architecture (PSSA) combines a selective recurrent state, hyperbolic episodic memory, online plasticity, and consolidation. Its checked-in memory reader computes a full softmax over all populated slots, while inference also computes a dense vocabulary projection for every token. We propose a dual-certified sparse inference method: Certified Sparse Retrieval (CSR) for episodic memory and Certified Vocabulary Projection (CVP) for the output head. CSR returns a support only when omitted softmax mass is certified below a user tolerance; CVP uses cluster upper bounds to compute exact greedy/top- $K$  results without evaluating every vocabulary row, and bounds the distributional error of temperature sampling. The paper proves value, logit, cross-entropy, argmax, gradient, and output-tail guarantees. At the default shape, memory-only CSR gives a  $1.125\times$  coordinate-work proxy, but a configuration with 32 output clusters and 64 exact rows reduces the output projection by  $21.33\times$  and raises the combined proxy to  $2.171\times$ , subject to certification and routing overhead. A deterministic checker exercises 7,712 theorem instances, including output bounds and exact argmax certificates. We also define evaluations beyond next-token loss: calibration, selective risk, episodic recall, continual-learning transfer, robustness, generation diversity, and latency/energy efficiency. These guarantees and metrics are distinguished from unmeasured claims about language quality.

## 1 Introduction

Transformer language models obtain content-based interaction by comparing tokens, but their per-step context processing grows with sequence length. PSSA explores a different architecture: a token-dependent state-space recurrence carries a fixed-size state, an episodic bank stores selected experiences, and plastic parameters adapt during a stream. The repository is implemented in Rust with scalar reference paths and staged CPU, CUDA, and WebGPU backends.

The project reports promising prototype measurements. On a parameter-matched approximately 1.5-million-parameter comparison, the README reports held-out cross-entropy 3.997 for PSSA versus 4.429 for a transformer and 200-token CPU generation times of 226 ms versus 2,735 ms [1]. Those results motivate an architectural performance contribution, but they do not prove that a sparse reader preserves the full reader’s behavior. In fact, the live method `retrieve_soft_into` at `src/memory.rs:270--324` scans all populated slots and aggregates all value coordinates.

A fixed top-4 replacement is not mathematically safe for every query. If the softmax is diffuse, the top four entries can contain very little probability mass. Likewise, a fixed output shortlist can miss the true argmax. The contribution of this paper is therefore dual certification: approximate only when computable bounds prove the requested fidelity, and fall back to exact evaluation otherwise. This also targets the output projection, which dominates the current coordinate-work model after memory retrieval is optimized.

## Contributions.

1. We define CSR, a metric-indexed, state-conditioned sparse memory read with an explicit error tolerance and an exact fallback.
2. We prove value, total-variation, logit, cross-entropy, argmax, and input-Jacobian error bounds for the sparse read.
3. We prove a metric-ball certificate using only triangle inequality and derive its conditional coordinate-work reduction.

4. We prove that the existing fast-to-slow adapter consolidation preserves the effective adapter exactly, so the proposed reader does not require changing that stability invariant.
5. We introduce CVP, prove its exact top- $K$  and bounded-sampling guarantees, and derive the combined complexity bound.
6. We provide reproducible certificate checks and an evaluation protocol covering calibration, memory, continual learning, robustness, generation, and systems efficiency.

The word “works” is used here in a precise sense: the method is guaranteed to meet a specified retrieval and downstream perturbation tolerance under stated assumptions, and is guaranteed to return the exact answer when it cannot certify sparsity. It does not mean that a mathematical error bound alone proves better language quality.

## 2 PSSA Background

### 2.1 Selective state-space core

Let  $e_t$  be a token embedding and  $x_t$  its affine RMS-normalized form. PSSA computes input-dependent parameters

$$\Delta_t = \text{softplus}(W_\Delta x_t), \quad (1)$$

$$B_t = W_B x_t, \quad C_t = W_C x_t, \quad (2)$$

$$A_{ij} = -\text{softplus}(A_{ij}^{\text{raw}}). \quad (3)$$

The diagonal discretization is

$$\bar{A}_{t,ij} = \exp(\Delta_{t,i} A_{ij}), \quad \bar{B}_{t,ij} = \Delta_{t,i} B_{t,j}, \quad (4)$$

$$h_{t,ij} = \bar{A}_{t,ij} h_{t-1,ij} + \bar{B}_{t,ij} x_{t,i}, \quad y_{t,i} = \sum_j C_{t,j} h_{t,ij}. \quad (5)$$

Because  $A_{ij} < 0$ , each transition is contractive in the scalar decay factor for positive  $\Delta_t$ . The learned timescales provide recurrent memory without rereading the full context, consistent with the broader selective-SSM literature [2, 3].

### 2.2 Current episodic reader

The query is conditioned on both the normalized token and the recurrent output:

$$q_t = W_{qx} x_t + W_{qh} y_t, \quad \tilde{q}_t = \frac{q_t}{1 + \|q_t\|}. \quad (6)$$

The projection lies in the open Poincare ball  $\mathbb{B}^k$ . For keys  $k_i \in \mathbb{B}^k$ , the code uses

$$d_{\mathbb{H}}(u, v) = 2 \operatorname{asinh} \left( \frac{\|u - v\|}{\sqrt{(1 - \|u\|^2)(1 - \|v\|^2)}} \right). \quad (7)$$

Let  $d_i = d_{\mathbb{H}}(\tilde{q}_t, k_i)$  and  $a_i = \exp(-d_i/\tau)$ . The exact reader is

$$p_i = \frac{a_i}{Z}, \quad Z = \sum_{i=1}^M a_i, \quad (8)$$

$$m = \sum_{i=1}^M p_i v_i. \quad (9)$$

The implementation computes each populated-slot distance, every exponential, and every value-coordinate accumulation. Thus its dominant coordinate work is  $\Theta(M(k + d))$ , not  $\Theta(Mk + Kd)$  for a fixed  $K$ .

## 2.3 Plasticity and consolidation

Memory injection is gated and projected before a rank- $r$  adapter and SiLU MLP:

$$g = \sigma(W_g x), \quad (10)$$

$$z = y/\sqrt{s} + g \odot W_p m + (U_f + U_s) \text{SiLU}(Dx), \quad (11)$$

$$z' = z + W_2 \text{SiLU}(W_1 z). \quad (12)$$

The memory write path uses novelty and a refractory confidence gate. The adapter consolidation implemented in `src/adapter.rs` is

$$U'_f = (1 - \alpha)U_f, \quad U'_s = U_s + \alpha U_f. \quad (13)$$

## 3 Certified Sparse Retrieval

### 3.1 Definitions

**Definition 3.1** (Tail mass and restricted read). For a support  $S \subseteq \{1, \dots, M\}$ , define its retained mass  $P_S = \sum_{i \in S} p_i$  and tail mass  $\delta_S = 1 - P_S$ . If  $P_S > 0$ , the restricted read is

$$\hat{m}_S = \sum_{i \in S} \hat{p}_i v_i, \quad \hat{p}_i = \frac{p_i}{P_S}. \quad (14)$$

**Definition 3.2** (Metric-ball index). Partition the memory keys into clusters  $C_1, \dots, C_J$ . Each cluster has a center  $c_j$  and radius  $r_j$  satisfying

$$d_{\mathbb{H}}(c_j, k_i) \leq r_j \quad \text{for all } i \in C_j. \quad (15)$$

For query  $q$ , define the lower distance bound

$$\ell_j(q) = \max\{0, d_{\mathbb{H}}(q, c_j) - r_j\}. \quad (16)$$

### 3.2 CSR algorithm

CSR orders clusters by increasing  $\ell_j(q)$ . It scans clusters in that order, computes exact distances for all keys in scanned clusters, and maintains the exact scanned weight

$$A_S = \sum_{i \in S} \exp(-d_i/\tau). \quad (17)$$

For unscanned clusters  $U$ , it computes the conservative upper weight

$$\bar{A}_U = \sum_{j \in U} |C_j| \exp(-\ell_j(q)/\tau), \quad \bar{\delta}_S = \frac{\bar{A}_U}{A_S + \bar{A}_U}. \quad (18)$$

The algorithm stops and returns the restricted read when  $\bar{\delta}_S \leq \varepsilon$ . If this condition is not met after all clusters are scanned, it returns the exact read. The current implementation uses a numerically shifted softmax; the same shift can be applied to both terms in Eq. (18).

This construction is state-conditioned because  $q = W_{qx}x + W_{qy}y$  includes the recurrent state output  $y$ . It is not a fixed token-to-memory lookup.

## 4 Formal Guarantees

We now prove that CSR is an error-bounded optimization. All norms below are Euclidean for vectors and induced spectral or Frobenius norms for matrices.

**Theorem 4.1** (Value retrieval error). *Assume  $\|v_i\| \leq V$  for every memory value. For any support  $S$  with tail mass  $\delta_S < 1$ ,*

$$\|m - \hat{m}_S\| \leq 2V\delta_S. \quad (19)$$

*The constant 2 is tight in general.*

*Proof.* Let  $P = P_S = 1 - \delta_S$  and define the omitted normalized mean  $m_{\bar{S}} = \delta_S^{-1} \sum_{i \notin S} p_i v_i$  when  $\delta_S > 0$ . Then

$$m = P\hat{m}_S + \delta_S m_{\bar{S}}, \quad m - \hat{m}_S = \delta_S(m_{\bar{S}} - \hat{m}_S).$$

Both means lie in the convex hull of vectors with norm at most  $V$ , so their norms are at most  $V$ . The triangle inequality gives Eq. (19). For tightness, take two groups with values  $v_i = Vu$  and  $v_i = -Vu$  for a unit vector  $u$ , with omitted mass  $\delta_S$  and retained mass  $1 - \delta_S$ . Then the difference has norm exactly  $2V\delta_S$ .  $\square$

**Corollary 4.2** (Optimal fixed-cardinality support). *Among all supports of cardinality  $K$ , the support containing the  $K$  largest probabilities minimizes the value bound in Eq. (19); it maximizes retained mass  $P_S$ .*

*Proof.* If a support contains  $i$  with  $p_i < p_j$  for an omitted  $j$ , swapping  $i$  for  $j$  increases retained mass by  $p_j - p_i$ . Repeating exchanges produces the top- $K$  support and cannot decrease  $P_S$ . Since the bound is monotone in  $\delta_S = 1 - P_S$ , it is optimal.  $\square$

**Theorem 4.3** (Metric-ball certification). *For any scanned support  $S$  and unscanned clusters  $U$ , the actual omitted mass satisfies*

$$\delta_S \leq \bar{\delta}_S = \frac{\bar{A}_U}{A_S + \bar{A}_U}. \quad (20)$$

Consequently, if CSR stops at  $\bar{\delta}_S \leq \varepsilon$ , then  $\|m - \hat{m}_S\| \leq 2V\varepsilon$ .

*Proof.* For any  $i \in C_j$ , the triangle inequality and Eq. (15) imply

$$d_{\mathbb{H}}(q, k_i) \geq d_{\mathbb{H}}(q, c_j) - d_{\mathbb{H}}(c_j, k_i) \geq d_{\mathbb{H}}(q, c_j) - r_j.$$

Distances are nonnegative, so  $d_i \geq \ell_j(q)$ . Therefore  $\exp(-d_i/\tau) \leq \exp(-\ell_j(q)/\tau)$  and the total unscanned weight is at most  $\bar{A}_U$ . The exact denominator is  $Z = A_S + A_U \geq A_S$ , with  $A_U \leq \bar{A}_U$ . Hence

$$\delta_S = \frac{A_U}{A_S + A_U} \leq \frac{\bar{A}_U}{A_S + \bar{A}_U}.$$

The value guarantee follows from Theorem 4.1.  $\square$

**Theorem 4.4** (Downstream logit and loss stability). *Let  $G = \text{diag}(g)$ , let  $R(z) = z + W_2 \text{SiLU}(W_1 z)$ , and let the logits be  $\ell(m) = W_o R(a + GW_p m)/\sqrt{d}$ . Define*

$$L_{\Phi} = \frac{\|W_o\|_2}{\sqrt{d}} (1 + \|W_2\|_2 L_{\text{SiLU}} \|W_1\|_2) \|GW_p\|_2, \quad (21)$$

where  $L_{\text{SiLU}} = \sup_x |\text{SiLU}'(x)|$ . If the support has tail mass  $\delta_S$ , then

$$\|\ell(m) - \ell(\hat{m}_S)\| \leq 2L_{\Phi} V \delta_S, \quad (22)$$

$$|\text{CE}(\ell(m), y) - \text{CE}(\ell(\hat{m}_S), y)| \leq 2\sqrt{2} L_{\Phi} V \delta_S. \quad (23)$$

If the full-logit top-two margin is  $\gamma = \ell_{(1)} - \ell_{(2)} > 4L_{\Phi} V \delta_S$ , the sparse and full readers predict the same token.

*Proof.* The SiLU function is globally Lipschitz with constant  $L_{\text{SiLU}} < \infty$ . The residual map satisfies

$$\|R(u) - R(v)\| \leq (1 + \|W_2\|_2 L_{\text{SiLU}} \|W_1\|_2) \|u - v\|$$

by the triangle inequality and induced-norm submultiplicativity. Applying the output matrix, memory gate, and Theorem 4.1 proves the logit inequality.

For one target  $y$ , cross-entropy has gradient  $\nabla_{\ell} \text{CE} = \text{softmax}(\ell) - e_y$ . Both terms are probability vectors, and  $\|\text{softmax}(\ell) - e_y\| \leq \sqrt{2}$ . The mean-value inequality gives the loss bound. Finally, if  $i^*$  is the full argmax, then  $\hat{\ell}_{i^*} \geq \ell_{i^*} - E$  and  $\hat{\ell}_j \leq \ell_j + E$  for every  $j$ , where  $E = 2L_{\Phi} V \delta_S$ . The top token is preserved when  $\gamma > 2E$ .  $\square$

**Theorem 4.5** (Input-gradient stability). *Assume  $\|v_i\| \leq V$  and  $\|\nabla_q d_{\mathbb{H}}(q, k_i)\| \leq G$  for every key. For the full and restricted memory maps,*

$$\|\nabla_q m - \nabla_q \hat{m}_S\|_F \leq \frac{6VG}{\tau} \delta_S. \quad (24)$$

*Proof.* Writing  $g_i = \nabla_q d_i$ , differentiation of the softmax gives

$$\nabla_q m = -\frac{1}{\tau} \sum_i p_i (v_i - m) g_i^\top.$$

Let  $a = \sum_i p_i g_i$  and  $b = \sum_i \hat{p}_i g_i$ . Expanding the difference as

$$\sum_i (p_i - \hat{p}_i) v_i g_i^\top - m a^\top + \hat{m}_S b^\top$$

and adding and subtracting  $\hat{m}_S a^\top$  gives three terms. The first is at most  $2VG\delta_S$ , the term  $(m - \hat{m}_S) a^\top$  is at most  $2VG\delta_S$  by Theorem 4.1, and  $\hat{m}_S (a - b)^\top$  is at most  $2VG\delta_S$  because the total variation distance between  $p$  and  $\hat{p}$  is  $\delta_S$ . Multiplication by  $1/\tau$  proves Eq. (24).  $\square$

**Theorem 4.6** (Exactness of fast-to-slow consolidation). *For any  $\alpha \in [0, 1]$ , Eq. (13) preserves the effective adapter:  $U'_f + U'_s = U_f + U_s$ . Therefore its adapter output and its input Jacobian are unchanged before any subsequent optimizer update.*

*Proof.* Direct substitution gives  $U'_f + U'_s = (1 - \alpha)U_f + U_s + \alpha U_f = U_f + U_s$ . The adapter output is a function of this sum and  $D$  only, so it and its derivative are identical.  $\square$

## 5 Certified Vocabulary Projection

The dense vocabulary head is a second bottleneck. Let the frozen output rows be  $w_i \in \mathbb{R}^d$  and let the latent vector entering the head be  $z$ . PSSA’s logits are

$$\ell_i(z) = \frac{w_i^\top z}{\sqrt{d}}, \quad i \in \{1, \dots, V\}. \quad (25)$$

Partition the vocabulary rows into clusters  $D_1, \dots, D_J$ . For each cluster store its centroid  $\mu_j$  and radius

$$\rho_j = \max_{i \in D_j} \|w_i - \mu_j\|. \quad (26)$$

The index is built after a checkpoint is loaded and can be rebuilt whenever the output rows change. For inference, it is read-only.

**Theorem 5.1** (Vocabulary cluster upper bound). *For every  $i \in D_j$ ,*

$$\ell_i(z) \leq u_j(z) := \frac{\mu_j^\top z + \rho_j \|z\|}{\sqrt{d}}. \quad (27)$$

*Proof.* Write  $w_i = \mu_j + (w_i - \mu_j)$ . By Cauchy–Schwarz,  $w_i^\top z \leq \mu_j^\top z + \|w_i - \mu_j\| \|z\| \leq \mu_j^\top z + \rho_j \|z\|$ . Divide by  $\sqrt{d}$ .  $\square$

**Exact greedy and top- $K$ .** CVP evaluates all  $J$  centroid scores and upper bounds, then visits clusters in decreasing  $u_j(z)$ . It computes exact row logits in visited clusters. For greedy decoding, it stops when the largest unvisited upper bound is no greater than the current best exact logit. For top- $K$ , it stops when that bound is no greater than the current exact  $K$ th logit. Ties are resolved by token ID and are retained in the candidate set when the upper bound equals the threshold.

**Theorem 5.2** (Exact branch-and-bound output). *Assume valid radii in Eq. (26). CVP’s greedy result and its top- $K$  result are identical to exhaustive evaluation, including the stated tie rule. If  $s$  rows are visited, its coordinate work is*

$$W_{\text{CVP}} = Jd + sd, \quad (28)$$

*apart from sorting and scalar bound operations.*

*Proof.* By Theorem 5.1, an unvisited row cannot exceed its cluster’s upper bound. If the largest unvisited bound is below the current best logit, no unvisited row can replace the current argmax. The top- $K$  case is identical: if the largest possible unvisited logit is below the current  $K$ th exact logit, no unvisited row can enter the top- $K$  set. Equality is retained to preserve tie semantics. The work consists of  $J$  centroid dot products and  $s$  exact row dot products.  $\square$

### Certified temperature sampling.

For temperature  $T > 0$ , let  $S$  be the visited rows and let  $A_S = \sum_{i \in S} \exp(\ell_i/T)$ . For unvisited clusters define

$$\bar{A}_U = \sum_{j \in U} |D_j| \exp(u_j/T), \quad \bar{\delta}_S^{\text{out}} = \frac{\bar{A}_U}{A_S + \bar{A}_U}. \quad (29)$$

**Theorem 5.3** (Output-distribution certificate). *The full temperature-softmax distribution  $p$  and the renormalized visited-row distribution  $\hat{p}_S$  satisfy*

$$\text{TV}(p, \hat{p}_S) = \sum_{i \notin S} p_i \leq \bar{\delta}_S^{\text{out}}. \quad (30)$$

Consequently, every event’s sampling probability changes by at most  $\bar{\delta}_S^{\text{out}}$ .

*Proof.* Theorem 5.1 gives  $\exp(\ell_i/T) \leq \exp(u_j/T)$  for every unvisited  $i \in D_j$ , so the unvisited unnormalized mass is at most  $\bar{A}_U$ . The proof of Theorem 4.3 then applies with logits in place of negative distances. The total-variation identity is the same renormalization identity used in the memory proof. For any event  $E$ ,  $|p(E) - \hat{p}_S(E)| \leq \text{TV}(p, \hat{p}_S)$ .  $\square$

## 6 Complexity Guarantees

Let  $M$  be the populated capacity,  $k$  the key width,  $d$  the value width, and  $s = |S|$  the number of scanned candidate keys. Full retrieval has coordinate work

$$W_{\text{full}} = M(k + d). \quad (31)$$

CSR with  $J$  cluster centers has coordinate work bounded by

$$W_{\text{CSR}} = Jk + s(k + d), \quad (32)$$

apart from sorting, scalar exponentials, index maintenance, and memory-system effects. This gives a conditional theorem:

**Proposition 6.1** (Certified work reduction). *If the certificate terminates with  $Jk + s(k + d) < M(k + d)$ , CSR performs strictly less coordinate work than the full reader while satisfying the value, logit, loss, and gradient guarantees above. If it does not terminate early, the exact fallback has identical output to the full reader.*

*Proof.* The inequality is the direct comparison of Eqs. (31) and (32). The guarantees follow from the certificate and previous theorems. Full fallback computes Eq. (9), hence is exact.  $\square$

At PSSA’s documented default  $M = 512$ ,  $k = 32$ ,  $d = 256$ , a full read has 147,456 coordinate units. A hypothetical exact top-4 aggregation after a full key scan has  $512 \cdot 32 + 4 \cdot 256 = 17,408$  units, an  $8.47\times$  retrieval reduction and a  $128\times$  reduction in value mixing. This is not a universal speedup: it is admissible only when the top-4 tail mass meets  $\varepsilon$ .

The output head costs  $Vd = 2048 \cdot 256 = 524,288$  coordinate units. With  $J = 32$  clusters and  $s = 64$  exact rows, CVP costs  $(J + s)d = 96 \cdot 256 = 24,576$  units, a  $21.33\times$  output reduction. If both certificates terminate on that configuration, the combined proxy improves from 1,167,360 dense units to 537,600 units, or  $2.171\times$ . This is a conditional analytical result; routing, sorting, cache behavior, fallback frequency, and checkpoint-index construction must be measured end-to-end.

## 7 Certificate Verification and Empirical Protocol

### 7.1 Deterministic verification

The script `research/verify_certificates.py` checks the theorem instances independently of Rust training. It performs 7,712 checks using deterministic binary64 arithmetic, including:

- retrieval error versus  $2V\delta$  on 80 random banks;
- total-variation equality  $\text{TV}(p, \hat{p}) = \delta$ ;
- exact top- $K$  support optimality by exhaustive enumeration;

Table 1: Dual coordinate-work proxy at  $V = 2048$ ,  $d = 256$ ,  $k = 32$ ,  $M = 512$ . The entries count multiply-accumulate coordinate pairs, not wall-clock FLOPs.

| Stage                        | Dense     | Memory CSR | Dual CSR+CVP |
|------------------------------|-----------|------------|--------------|
| Selective projections        | 73,728    | 73,728     | 73,728       |
| Query projections            | 16,384    | 16,384     | 16,384       |
| SSM state update             | 4,096     | 4,096      | 4,096        |
| Memory retrieval             | 147,456   | 17,408     | 17,408       |
| Memory gate and projection   | 131,072   | 131,072    | 131,072      |
| Rank-16 adapter              | 8,192     | 8,192      | 8,192        |
| SiLU MLP                     | 262,144   | 262,144    | 262,144      |
| Vocabulary output projection | 524,288   | 524,288    | 24,576       |
| Total                        | 1,167,360 | 1,037,312  | 537,600      |
| Ratio                        | 1.000     | 1.125      | 2.171        |

- metric-ball certificates on random clustered banks;
- finite-difference checks of the hyperbolic-distance Jacobian and memory Jacobian; and
- downstream logit, cross-entropy, and certified-argmax bounds.
- vocabulary-row upper bounds, exact branch-and-bound argmax, and temperature-softmax tail certificates.

The checker also includes a tight family that demonstrates why unconditional top-4 retrieval is invalid. Four close keys receive probability mass while 508 distractor keys are separated by a hyperbolic distance gap  $g$ . At temperature  $\tau$ , the top-4 tail mass is exactly

$$\delta_4 = \frac{(M - 4) \exp(-g/\tau)}{4 + (M - 4) \exp(-g/\tau)}. \quad (33)$$

With values  $+Vu$  on the retained keys and  $-Vu$  on distractors, the value error is exactly  $2V\delta_4$ , meeting Theorem 4.1. For  $M = 512$  and  $\varepsilon = 0.01$ , the checker finds that a gap of  $8\tau$  still needs 388 keys under this adversarial family, while a gap of  $10\tau$  makes four keys sufficient. Thus the certified adaptive support is the mathematically justified optimization; fixed top-4 is only a special case when its certificate succeeds.

In a separated one-dimensional metric fixture with 512 keys and nine clusters, CSR scans four keys after evaluating nine centers, for 13 key-distance evaluations. Its actual omitted mass is  $7.82e - 05$  and the certificate is 0.00145. This demonstrates the intended fast path while preserving the exact fallback for diffuse or poorly indexed banks.

For the output head, the checker validates 5,201 row upper bounds, exact branch-and-bound argmax cases, and sampling-tail certificates. In the separated  $V = 2048$ ,  $d = 256$  fixture, CVP evaluates 32 cluster centroids and 64 exact rows rather than all 2,048 rows. Its output projection is therefore  $21.33\times$  smaller in coordinate work, and the dual memory-plus-output proxy is  $2.171\times$ .

## 7.2 Required PSSA experiments

The mathematical results do not establish language quality. After implementing CSR in a separate experimental branch, compare:

1. C0: current full-bank softmax;
2. C1: exact top- $K$  support after a full distance scan;
3. C2: CSR with metric clusters and exact fallback;
4. C3: CSR with adaptive  $\varepsilon$  or  $K$  based on read entropy; and
5. C4: C3 plus CVP for exact greedy/top- $K$  or certified sampling; and
6. C5: C4 plus surprise-aware writes and the existing consolidation.

Use at least five paired seeds, the same tokenizer and optimizer schedule, and report retrieval time, end-to-end tokens/s, held-out cross-entropy, perplexity, next-token accuracy, fixed-prompt generations, fact recall, retrieval recall, and task-A loss after task-B training. A practical acceptance criterion is a measured  $1.5\times$  end-to-end inference improvement at  $M = 512$ , no more than 0.05 nats held-out degradation, no significant fact-recall loss, and preserved task-A retention. These are preregistered criteria, not results claimed here.

## 8 Mathematical Kernel Transformations

The speed target is not limited to reducing asymptotic model work. Several kernel transformations preserve PSSA’s defined computation exactly and reduce launches, memory traffic, or parallel span. The full implementation plan is in `research/kernel_performance.md`.

### 8.1 Packed exact projections

For token rows  $X$  and compatible projections, define

$$W_{\oplus} = \begin{bmatrix} W_{\Delta} \\ W_B \\ W_C \end{bmatrix}. \quad (34)$$

Then

$$XW_{\oplus}^{\top} = [XW_{\Delta}^{\top} \mid XW_B^{\top} \mid XW_C^{\top}]. \quad (35)$$

Splitting the rows recovers all three original projections exactly. The same identity applies to shared-input gate, adapter-down, and query projections when their dependency stage matches. This reduces GEMM launches and repeated input reads without changing scalar multiply-add count or the model function.

### 8.2 Associative recurrence and loss reductions

Every diagonal SSM coordinate has the affine map  $h_t = a_t h_{t-1} + b_t$ . Its composition

$$(a_2, b_2) \otimes (a_1, b_1) = (a_2 a_1, a_2 b_1 + b_2) \quad (36)$$

is associative, so a segmented prefix scan has work  $O(BLd_m s)$  and span  $O(\log L)$  for  $B$  independent sequences, while preserving document boundaries. A tiled scan uses span  $O(T + \log(L/T))$  for short chunks. The backward recurrence uses the reverse monoid.

The same idea applies to the vocabulary loss. A log-sum-exp tile is summarized by  $(m, s)$  where  $m = \max_i x_i$  and  $s = \sum_i \exp(x_i - m)$ . The associative combine operation is

$$(m_1, s_1) \oplus (m_2, s_2) = (m, s_1 e^{m_1 - m} + s_2 e^{m_2 - m}), \quad m = \max(m_1, m_2). \quad (37)$$

Thus a fused output-loss kernel can avoid materializing the full probability matrix while preserving the flat-softmax objective. These are exact algebraic kernel changes; numerical reassociation remains subject to the selected accumulation precision.

### 8.3 Certified numerical kernels

For quantized operands  $W_q = W + E_W$  and  $x_q = x + e_x$ ,

$$\|W_q x_q - Wx\| \leq \|E_W\|_2 \|x\|_2 + \|W + E_W\|_2 \|e_x\|_2. \quad (38)$$

The existing PSSA Lipschitz bound then gives  $\|\Delta\ell\| \leq L_{\Phi} \|\Delta z\|$  and  $|\Delta \text{CE}| \leq \sqrt{2} \|\Delta\ell\|$ . Inference may use BF16/FP16 or tensor-core products only when the bound is below a declared tolerance; training can retain FP32 master weights and moments, accumulate in FP32, and recompute out-of-budget batches. A truncated SVD has the analogous exact operator bound

$$\|(W - W_r)x\|_2 \leq \sigma_{r+1}(W) \|x\|_2. \quad (39)$$

Low-rank kernels are therefore an explicitly budgeted model variant, whereas packing, scans, and online log-sum-exp are exact transformations.

## 9 Evaluation Beyond Next-Token Loss

Next-token cross-entropy is a useful average score but does not test calibration, episodic recall, forgetting, robustness, or deployment efficiency. We define a multi-axis protocol in `research/evaluation_protocol.md`; the dependency-free implementation is `research/evaluation_metrics.py`.

### 9.1 Calibration and selective prediction

For probability vector  $p_t$ , target  $y_t$ , and predicted confidence  $c_t = \max_v p_{t,v}$ , report the multiclass Brier score

$$\text{Brier} = \frac{1}{N} \sum_{t=1}^N \sum_v (p_{t,v} - \mathbf{1}[v = y_t])^2. \quad (40)$$

For confidence bins  $B_b$ , expected calibration error is

$$\text{ECE} = \sum_b \frac{|B_b|}{N} \left| \frac{1}{|B_b|} \sum_{t \in B_b} \mathbf{1}[\hat{y}_t = y_t] - \frac{1}{|B_b|} \sum_{t \in B_b} c_t \right|. \quad (41)$$

Risk-coverage curves sort predictions by confidence and measure error after retaining only the top-confidence fraction. Report risk at 50%, 75%, 90%, and 100% coverage and the area under this curve. This directly tests whether CSR’s uncertainty-dependent fallback is meaningful.

### 9.2 Memory, continual learning, and behavior

The full memory reader is an oracle for retrieval experiments. Report `recall@1`, `recall@4`, `precision@K`, mean reciprocal rank, actual omitted mass, the certified upper bound, observed value/logit/loss error, write precision, eviction count, age-weighted recall, and contradiction resistance. For a task loss matrix  $L_{a,t}$ , report

$$\text{Forget}_a = L_{a,T} - \min_{t \geq a} L_{a,t}, \quad (42)$$

$$\text{BWT}_a = L_{a,T} - L_{a,a}, \quad (43)$$

$$\text{FWT}_a = L_{a+1,\text{history}} - L_{a+1,\text{scratch}}. \quad (44)$$

Positive loss change denotes forgetting; negative transfer is an improvement. The protocol also measures fact recall under distractor gaps, order changes, repeated keys, token deletion, paraphrase, and counterfactual updates.

For generated sequences, retain every prompt and seed and report exact reference continuation or fact-answer accuracy where available, distinct- $n$ , repeated- $n$ -gram rate, longest repeated span, normalized log-probability, and prompt consistency across seeds. Human ratings may be secondary evidence but must not replace the deterministic metrics.

### 9.3 Systems efficiency

Report prefill and decode p50/p95 latency, tokens per second, peak resident memory, index-build time, fallback rate, and energy per token. For training, include optimizer updates and target-token exposure. The most useful deployment metric is time-to-quality,

$$\text{TTQ}(q) = \min_t \{t : \text{metric}(t) \geq q\}, \quad (45)$$

which prevents a fast but low-quality method from appearing superior through kernel timing alone. The metric sanity artifact contains a Brier score of 0.36585, ECE of 0.345, risk-coverage AUC of 0.14583, memory-bound validation, continual-loss changes, diversity scores, and latency/energy summaries for a deterministic fixture; these values demonstrate the implementation, not PSSA performance.

## 10 Related Work

Structured state-space models make long sequence computation efficient through linear recurrences and structured parameterizations [2]; selective SSMS condition the transition on input content and report fast inference [3]. PSSA

follows this family while adding a writable episodic store. Hyperbolic embeddings provide a geometry in which radial position can encode hierarchy and similarity [4]. Sparsemax demonstrates that sparse probability supports can be useful for attention [5], while top- $K$  attention studies approximate dense attention by retaining high-scoring entries [6]. CSR differs in its guarantee: it does not assume a fixed support is accurate; it certifies omitted softmax mass using metric bounds and uses exact fallback. The protected write and consolidation motivations are related to external-memory models [7], low-rank adaptation [8], and continual-learning stability [9], but the theorem here concerns PSSA’s specific state-conditioned hyperbolic memory read.

## 11 Limitations

The value and downstream bounds require bounded values, and the gradient bound requires bounded hyperbolic-distance derivatives. Poincare derivatives can become large near the boundary, so a practical implementation must track or conservatively bound  $V$  and  $G$ , or use the exact fallback. Cluster radii must be valid; stale or underestimated radii invalidate the certificate. Building and maintaining an index costs time and memory. Coordinate work is not a hardware timing model, and a local retrieval reduction may be hidden by output projection, cache misses, or backend transfers.

Most importantly, the theorems establish controlled approximation, not improved semantics. Better intelligence requires training experiments on held-out and continual-learning tasks. The current paper reports no new PSSA language-model score and does not alter production code.

## 12 Conclusion

PSSA provides a promising architecture for persistent sequence processing, but its current memory reader pays for all stored values and its output head scores every vocabulary row on every token. Dual-certified sparse inference addresses both bottlenecks: CSR bounds episodic omission and CVP bounds vocabulary omission, while exact fallbacks preserve correctness when the data do not permit sparsity.

The method is mathematically useful precisely because it refuses to claim a speedup when the data do not permit one. At the default dimensions, memory-only CSR improves the coordinate proxy by  $1.125\times$ , while the dual configuration with 32 output clusters and 64 exact rows reaches  $2.171\times$  under its certificate conditions. The next contribution should be a measured implementation of CSR and CVP, with calibration, retrieval, continual-learning, robustness, generation, and systems metrics connecting speed to intelligence and fidelity rather than relying on next-token loss alone.

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